

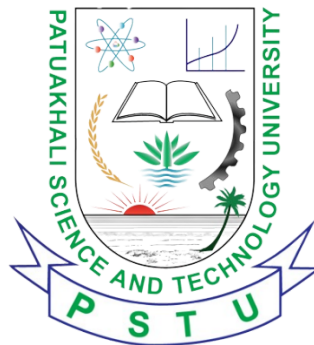
Project and Thesis Automation System

Using Next.js, Express.js and PostgreSQL

by

Arfan Ahmed

PROJECT SUBMITTED IN PARTIAL FULLFILLMENT OF THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING



FACULTY OF COMPUTER SCIENCE & ENGINEERING
PATUAKHALI SCIENCE & TECHNOLOGY UNIVERSITY

2th February, 2026

DECLARATION OF ORIGINAL WORK

The project titled “Project and Thesis Automation System Using Next.js and Express.js Framework and PostgreSQL”, submitted by Arfan Ahmed, Roll No. **2002040**, Session: 2020-21 to the Faculty of Computer Science and Engineering, Patuakhali Science & Technology University, has been accepted as satisfactory for the fulfilment of the requirements for the degree of Bachelor of Science in Computer Science & Engineering and approved as to its style and contents.

Board of Examinee

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Name of Internal Member : Professor Dr. Abdul Masud

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Certificate

This is to certify that this is a record of the project presented by the student whose name are given below in fulfillment of the requirements of the degree of Bachelor of Science in Computer Science and Engineering.

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Patuakhali Science & Technology University

DEDICATION

We dedicate this project to Allah Almighty, our Creator, our strong pillar, and our source of inspiration, wisdom, knowledge, and understanding. We also dedicate this work to our family and friends, whose support and encouragement have been invaluable throughout this journey.

Letter of Approval

This Project Submitted by Arfan Ahmed bearing ID No. 2002040 in partial fulfillment of Final Project Submission for B.Sc. in CSE degree has been examined and accepted for further process.

Approved

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Professor Chinmay Bepery

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ABSTRACT

The Project and Thesis Automation System is designed to streamline the project proposal submission and supervision management at Patuakhali Science and Technology University (PSTU). Built using Next.js, Node.js, Prisma ORM, and PostgreSQL, the system provides secure, role-based access for admins, supervisors, and students. It automates tasks like project proposal submission, supervisor allocation, task assignment, and real-time progress tracking. The system also enforces strict security through OTP-based authentication and ensures seamless communication via an automated email notification system for proposal approvals and rejections. This platform enhances the academic workflow by eliminating manual paper-based processes and reducing communication gaps between students and teachers. Supervisors can easily review proposals, accept or reject them via email triggers, and assign tasks, while students can access their approval status and submit project files securely via cloud storage. Admins have complete oversight of system operations, including managing course teachers and maintaining user records. The system's user-friendly design and robust architecture make it a reliable tool for the department, ensuring transparency and supporting future scalability for additional features.

ACKNOWLEDGMENTS

First, I am thankful to almighty Allah. He blesses me to complete this project successfully and deliver this within deadline.

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of people whose ceaseless cooperation made it possible, whose constant guidance and encouragement crown all efforts with success.

I am also grateful to our supervisor Professor Chinmay Bepery, Professor, Department of Computer Science and Information Technology, Faculty of Computer Science & Engineering, Patuakhali Science & Technology University.

I am also grateful to CSE 18th batch, Patuakhali Science & Technology University.

With Best Regards,

Arfan Ahmed

DECLARATION

We declare that the work presented in this “Project and Thesis Automation System Using Next.js and Express.js Framework and PostgreSQL” submitted to the Patuakhali Science and Technology University, for the award of the Bachelor of Science in Computer Science and Engineering degree, is our original work. We have not plagiarized or submitted the same work for the award of any other degree. In-case this undertaking is found incorrect, I accept that my degree may be unconditionally withdrawn.

.....
Arfan Ahmed

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CHAPTER 1

INTRODUCTION

1.1 Introduction

This report talks about how we made the Project and Thesis Automation System and how it works, and it is meant to make things easier and less complicated for the academic research and the supervision work at Patuakhali Science and Technology University which is also known as PSTU. The system was built using the Next.js framework and Node.js and Express.js and it uses Prisma ORM for managing the database with PostgreSQL, so the interface is very easy to use and it is responsive, so it works on many screens and it was designed using Tailwind CSS and Shadcn UI and TypeScript.

The system has many different parts for the administration and for the school work such as sending in project proposals online using the computer and picking supervisors and giving out tasks and keeping files safe and secure. It sends emails automatically when a proposal is accepted or when it is rejected, which is how people get messages in their inbox, and it follows the rules of the university for the thesis guidelines, and it also uses OTP based security for logging in. The platform lets the teachers manage the student groups and they can see how much work the students have done, and they can check tasks easily and at the same time the students can update what they are doing from a distance or remotely.

1.2 Motivation

In the contemporary academic landscape, the imperative for a robust research and project automation system is paramount. With the escalating focus on research and development in higher education, the absence of a centralized platform for managing student theses and projects poses formidable challenges for the department. Without an efficient web-based system, students grapple with the intricate orchestration of proposal submissions, finding available supervisors, and tracking their progress manually. Moreover, the lack of a unified repository for project files and feedback impedes students' ability to communicate effectively with their supervisors, often leading to delays and miscommunication.

The development of the Project and Thesis Automation System using Next.js, Node.js, and PostgreSQL is spurred by a commitment to surmount these obstacles, furnishing the department with a comprehensive solution. By centralizing the proposal submission process, automating supervisor allocation, and integrating secure Email Notifications, the system promises streamlined workflows, alleviated administrative burdens, and enhanced student-supervisor collaboration. This initiative aims to replace the traditional paper-based methods with a modern,

transparent digital environment, thereby fostering an atmosphere conducive to academic excellence and institutional efficiency.

1.3 Existing System

Patuakhali Science and Technology University (PSTU) currently employs a manual system for managing final year projects and thesis works. This system relies on paper-based proposal submissions, frequent in-person interactions, and manual record-keeping for processing approvals and task assignments.

Here's a breakdown of the existing project management process:

- **Proposal Submission:** Students prepare project or thesis proposals physically, print them, and submit hard copies to potential supervisors by visiting their offices personally.
- **Processing:** Faculty members manually review the submitted physical documents to check the feasibility of the topic and availability of slots.
- **Selection & Notification:** Supervisor selection and proposal approval decisions are made based on manual evaluation. Accepted students are notified verbally or through physical notice boards, which often leads to delays.
- **Task Management:** Upon acceptance, task assignments and progress updates are handled manually or via scattered emails, without a centralized log of activities.

This manual system presents several limitations:

- **Inefficiency:** The paper-based process is time-consuming for both students (who have to run after teachers) and supervisors (who have to manage piles of papers).
- **Data Loss & Errors:** Manual record-keeping increases the risk of losing proposal documents and creates inconsistencies in tracking student progress.
- **Communication Gap:** Students, especially those working remotely or off-campus, face difficulties in getting timely feedback and scheduling consultations with supervisors.

The existing manual project and thesis management system at PSTU hinders efficiency, transparency, and accessibility in the research workflow. The need for a dedicated, software-based Project and Thesis Automation System is evident to address these shortcomings.

1.4 Updated System

The updated Project and Thesis Automation System introduce a fully integrated module designed to streamline project proposal submission, supervisor allocation, and research progress management. It provides students with intuitive interfaces for viewing available supervisors, submitting project proposals, and uploading thesis documents via secure cloud storage.

Administrators can efficiently manage and update teacher lists, monitor proposal activities, and generate student group reports.

The system includes secure OTP-based authentication, adherence to university thesis guidelines, and handling of file submissions via Cloud nary to ensure data safety. A real-time email notification system facilitates smooth communication, ensuring instant alerts for proposal approvals or rejections. Automated notifications remind students of pending tasks and communicate status updates regarding their research progress.

Designed with a responsive interface (using Next.js and Tailwind CSS), the system is accessible from both desktop and mobile devices, offering users a seamless experience across platforms. This enhanced version improves academic project management, fostering greater efficiency and transparency for both students and university faculty members.

1.5 Contributions

The Project and Thesis Automation System was made using the Next.js framework and also PostgreSQL and it adds a lot of good things to the area of the academic research management and research administration. This system uses Next.js, its strong server side rendering and also its static site generation so it gives a solution that is cohesive and it can grow bigger for making the proposal submissions easier and better and it also makes the communication inside the department much better as well.

At the middle of it the system follows a Component Based Architecture through React and this helps with the separation of concerns, so it allows for a clear split between the user interface and the API logic and the data management layers too. This clear way of building things is helped by the TypeScript, and it makes the code easier to keep working on and it makes the types safe so it creates a solid foundation for the future when more things are added later.

Also the use of the Prisma ORM or Object Relational Mapping makes the database work with the PostgreSQL much easier and smoother, so it means we don't need to write hard manual SQL queries by hand and it lowers the danger of the SQL injection problems. This makes sure data management is very strong and safe which is very important for protecting the secret research data of students and also the feedback from the supervisors.

Putting in a custom OTP based login system gives the admins the power to set up specific access controls, and it helps with the role-based permissions for the Student and the Supervisor and the Admin and it keeps the data private. Also the use of the Tailwind CSS lets us make the user interfaces that are dynamic and responsive and easy to use so it makes the experience for the user much better on desktop computers and on mobile phones too.

1.6 Objectives

The updated Project and Thesis Automation System has several main goals that want to make the current way of doing things by hand more efficient and clearer and safe as well.

Make the proposal submission fully automatic so that the whole project and thesis process is easy and smooth also it includes picking the supervisor and sending the topic and uploading the digital files so there is less work for people to do by hand and less paper too.

Improve the way the users log in and make the security stronger by using OTP based logins for the admins and the supervisors and the students as well so that only the right people can get in and keep the research data safe and protected.

Help everyone talk to each other in real time and put in an automatic email system that sends alerts right away when a proposal is approved or rejected or when tasks are given out so the gap in the communication is fixed and people can talk better.

Make data management and the way data is handled better by using PostgreSQL and Prisma ORM to fix the database schema which helps with student groups and files and feedback from the supervisors, so it works well and there is no extra data that is not needed.

Make the user experience better for the people using it and make sure the interface is easy to use and looks good on the computers and on the phones so people can track their progress while they are moving around and on the go.

Plan for more things in the future and build a system that can change and grow later on so we can add things like automatic schedules for the defense and checking for copied work and digital grading too.

1.7 Organization of Project Reports

Chapter 2 Shows the literature review of the project.

Chapter 3 Shows the Methodology of the Project.

Chapter 4 The design of the database described.

Chapter 5 Software Design Process Described.

Chapter 6 This chapter introduces the Conclusion, the limitations of the project and prospective future scopes.

1.8 Summary

This introduction chapter gives a big look and a full summary of the Project and Thesis Automation System and its main goal and purpose, so it explains the history and the reasons for moving from a manual paper-based way to a digital framework and this explains the transition which is moving from one thing to another. It also shows the systems specific scope and the key goals and objectives which are the things we want to do. This part explains the way we worked, and the methodology used in the building phase, and it also tells how the rest of the chapters are set up and organized in the report.

CHAPTER 2

LITERATURE REVIEW OF PROJECT

2.1 Introduction

This chapter talks about the literature review of the Project and Thesis Automation System, and it also includes the definition of what a literature review is and what it means at the start of the chapter. It looks at the other similar academic project management systems that are already out there, and it also does a look at how they compare our proposed system to the ones we reviewed so we can see how they are the same and how they are different between them. This part of the report helps to build the theoretical and practical foundation for making the thesis supervision process automatic and it establishes the start of the work so the process can be done by a machine and not manually.

2.2 Literature Review

A literature review is basically a deep look and a detailed study of the books and papers about a specific research topic that is being studied and it can be a big part of the whole research process, or it can also stand by itself as its own separate project that is not part of something else. This literature review is going to look at the systems that are already out there right now which share similarities and are like our Project and Thesis Automation System because they are similar in how they work.

2.3 Similar System

The websites biis.buet.ac.bd, sust.edu, kuet.ac.bd and ru.ac.bd are web-based software programs and they are made and designed to handle and manage the academic research and the projects for the school and university and there are several systems that are based on that same thing. We are going to look at and compare these websites and sites based on our new updated version and the future we have for the later time.

2.4 Comparisons with our System

When we look at the other university's project management systems, our Project and Thesis Automation System has similar ways of doing things for the students and the proposals and the supervisors and the tasks as well. Our student model gets all the important details like the student ID and the registration ID and also the academic affiliation which is where they belong, and the proposal model handles the project details such as the title and description and the domain too. In a similar way, our supervisor model keeps the faculty information inside it, and the task model puts the academic project milestones in order so they are organized. Both of these systems probably have the same basic parts but there might be some different things in the specific features and the ways they work which are made for each university's thesis supervision process and the academic structure. Even though there are different changes, the main goal of managing the student projects and the research info stays the same and does not change across both systems.

2.5 Summary

This part talks about looking at the existing literature and the books that were written before and also similar apps for the Project and Thesis Automation System and also a comparison is shown to compare our system with those other apps and tools.

CHAPTER 3

METHODOLOGY

3.1 Introduction of Methodology

The methodology part explains the step by step way we used to make and build the Project and Thesis Automation System and it talks about the research plan and how we got the data and the system building steps and also any problems we ran into or challenges we faced.

The methodology is very important for making sure the project results are right and correct and also reliable so people can trust them and by using a organized research plan and the right tools for building the project aims to reach the goals well and in an effective way. This part shows how every step of the project was planned and done and checked to give a system that works and is reliable and this ensures the process is clear and honest and has transparency for everyone to see.

3.2 Facts finding techniques

Fact finding techniques are the ways and methods used to gather info and data about a specific system or a problem so that we can analyze it and understand it better. For this project the fact-finding techniques we used include interviews with the CSE faculty members and the Dean office staff plus we did surveys of the CSE students and we watched the current manual thesis submission steps. We also looked at the existing departmental guidelines and the reports and these techniques help to find out the requirements and the needs and the constraints and the chances for the system and this helps the design and the building work.

3.3 Software Development Life Cycle (SDLC)

The Software Development Life Cycle which is called SDLC, gives a plan and some rules for making good software that works well and when we use the SDLC the main goal is to build a system that meets what the primary owners' expectations are. In the case of our Project and Thesis Automation System project we are following the SDLC to ensure that our application is reliable and meets the needs of the CSE faculty and the students. By putting the SDLC into our development work we want to make a high-quality system that is fast and does the job and is easy for the people to use.

3.3.1 Software process model

The software model we used for the Project and Thesis Automation System is the Waterfall Model and it is a very simple way to make software. This model gives a structured and also a sequential approach for development and it has different parts like requirement analysis and the design and the implementation and the testing and also the maintenance. For the Project and

Thesis Automation System each part moves in a straight line and one phase must be finished and completed before the next one starts up. The Waterfall Model makes sure the development is systematic and organized and it helps make things clear and easy to predict during the whole project life cycle. By using this model the Project and Thesis Automation System wants to get all the requirements together and make a strong system design and do the work carefully and test it well so it is a reliable and efficient system for the CSE Faculty.

3.3.2 Feasibility Study

The feasibility study is very important and also a key phase for the development of the Project and Thesis Automation System. It involves a big and a wide analysis of the technical and the economic and the operational and the scheduling factors to see if the system is possible and if it is practical for us to use. The study showed that the system is feasible and it can be done and the benefits are more than the costs so it is worth the money. The Project and Thesis Automation System will make the department's management better and it will make things more efficient and more effective for the proposal submission and the thesis supervision tasks. This assessment shows the value of the Project and Thesis Automation System for the CSE Faculty and it is a good asset for making the workflow better and making the research experience much better for everyone.

3.3.3 Technical Feasibility

- Compatibility of the proposed system with the existing IT infrastructure of the CSE Faculty.
- Having all the necessary hardware and software like Nextjs and PostgreSQL for the development and for the testing and for the deployment of the system.
- Good network infrastructure to help with the system real time operations and for the cloud-based file uploads which go to the cloud.
- The way it can handle more thesis data and handle many users at the same time when the deadlines are coming up and the dates are close.
- The security and privacy steps like OTP and role-based access to make sure the sensitive research info is safe and protected.

3.3.4 Operational Feasibility

Checking if the Project and Thesis Automation System works well shows that its efficient and it works good in the CSE Faculty and some of the main points that show it works well are below.

- User Friendly Design: The system has a simple interface so its easy to use for students and teachers and so they don't need much training at all.
- Strong Testing: We did a lot of testing to make sure the system is strong and robust and it works without problems even when many people are turning things in.

- Growing the System: The system is built so it can grow larger in the future and this means it can change for the new research needs and also more users.
- Getting into the System: Because its web based it means anyone can use it from anywhere and also at any time and this makes it very easy and flexible for everyone.

3.3.5. Economic Feasibility

The system will also help in making the cost of the paper work and the physical storage less because most of the project proposals and the thesis papers will be stored in a digital way on the computer and kept there and the system will make the administrative resources for the faculty better as it will make the work of looking over students easier and smoother which increases how fast the proposal approval and the tracking process happens.

3.4 Summary

The way we are making the system is told in this chapter where we have picked the waterfall way of doing things because it works well to make sure things are done in a set order and one after another. The SDLC will be followed in the development of our system and we will follow the system development life cycle which is the plan for making the software.

CHAPTER 4

DATABASE DESIGN

4.1 Database Design

The database design is really important and it is a big part of the Project and Thesis Automation System because it is about organizing and structuring how the data is kept and stored so that it can be managed and retrieved back again. The team will use modern database management systems and tools that are new so we can make sure the performance is good and it can grow bigger as well, and keeping the data safe and making sure it is correct will be a very big deal. We will use strict rules for who can see the the data and we will make copies of the data for backups so the data is safe and the design will focus on making the data organized in a logical way that makes sense. This helps with moving and managing the data throughout the whole systems life cycle until the end when it is finished.

4.2 Entity Relationship Diagram

Entity Relationship Diagram (ERD) is a picture and graphical representation of the things called entities and how they relate to each other in a database. It helps people to see the database structure with their eyes and it makes sure that all the data is organized and put in the right way.

4.2.1 Project and Thesis Automation System Using Next.js Express.js Framework and PostgreSQL ER Diagram

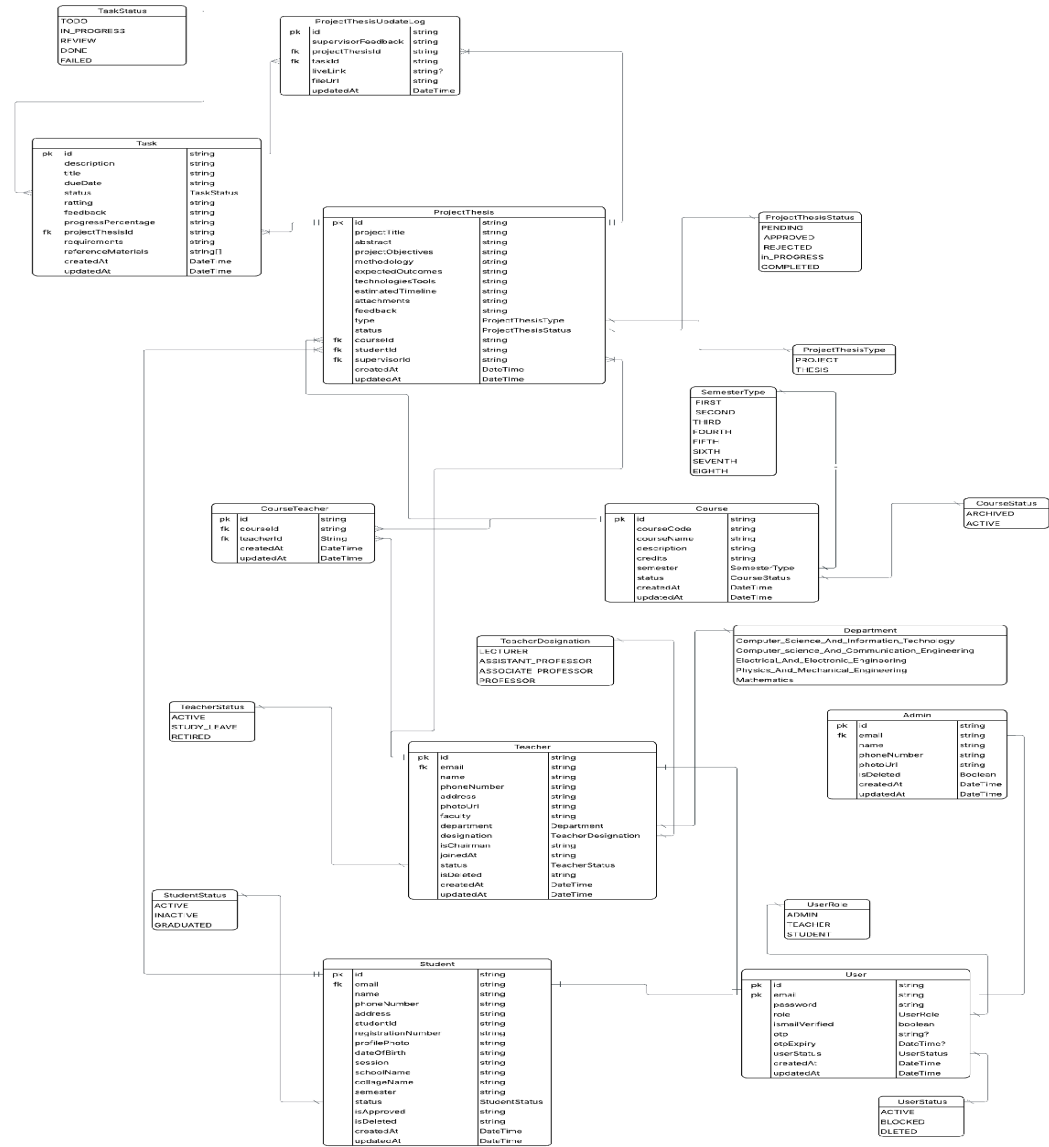


Figure 1: Project and Thesis Automation System ER Diagram

4.3 Summary

This chapter presented the database design approach used for the Project and Thesis Automation System, built on PostgreSQL and managed through the Prisma ORM. The Entity Relationship Diagram illustrated the core entities of the system, including Student, Teacher, Admin, Course, CourseTeacher, ProjectThesis, and Task, along with their attributes and the relationships that connect them. This design ensures that student proposals, supervisor assignments, task tracking, and course information are stored in a structured and consistent manner that supports the system's day-to-day operations.

CHAPTER 5

SOFTWARE DESIGN

5.1 Introduction

In the software design part of the Project and Thesis Automation System Using Next.js & Express.js Framework and PostgreSQL we will change the needs into a software picture and this includes the parts and the interfaces and the data that is needed for the implementation. The design will show the way for the development team and it will write down the architecture and the needs and the design choices to make sure everything matches the project rules and the industry standards and the rules of the industry.

5.2 Overall design

5.2.1 Admin Panel

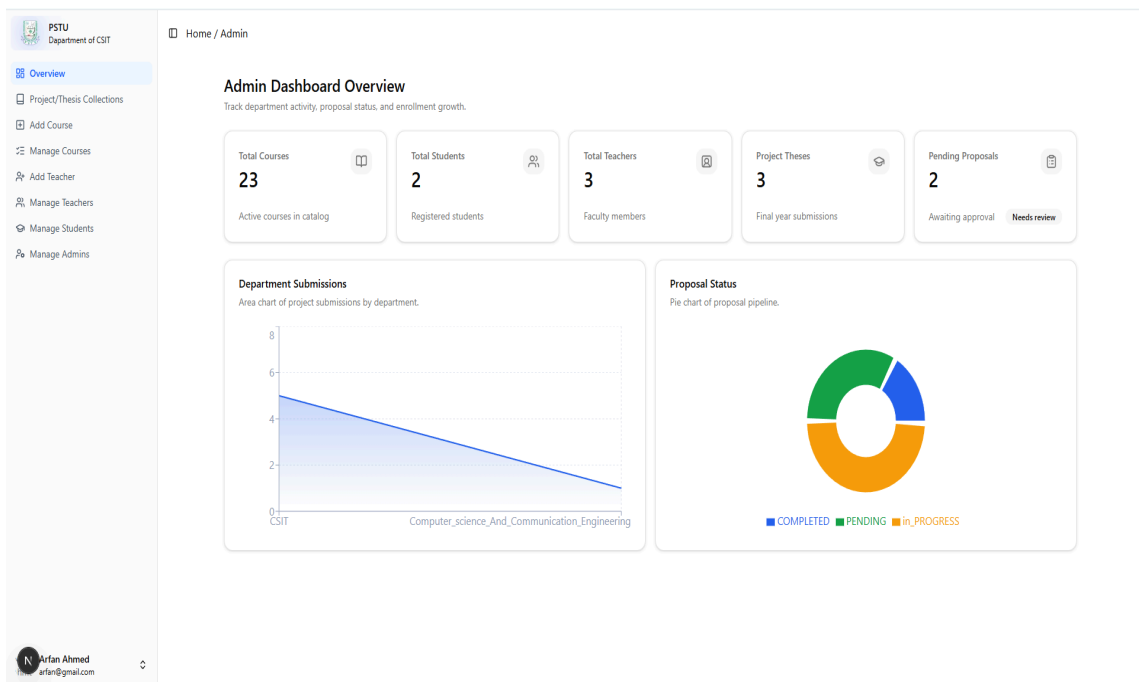


Figure 2: Admin Panel

5.2.2 Teacher Admin Pannel

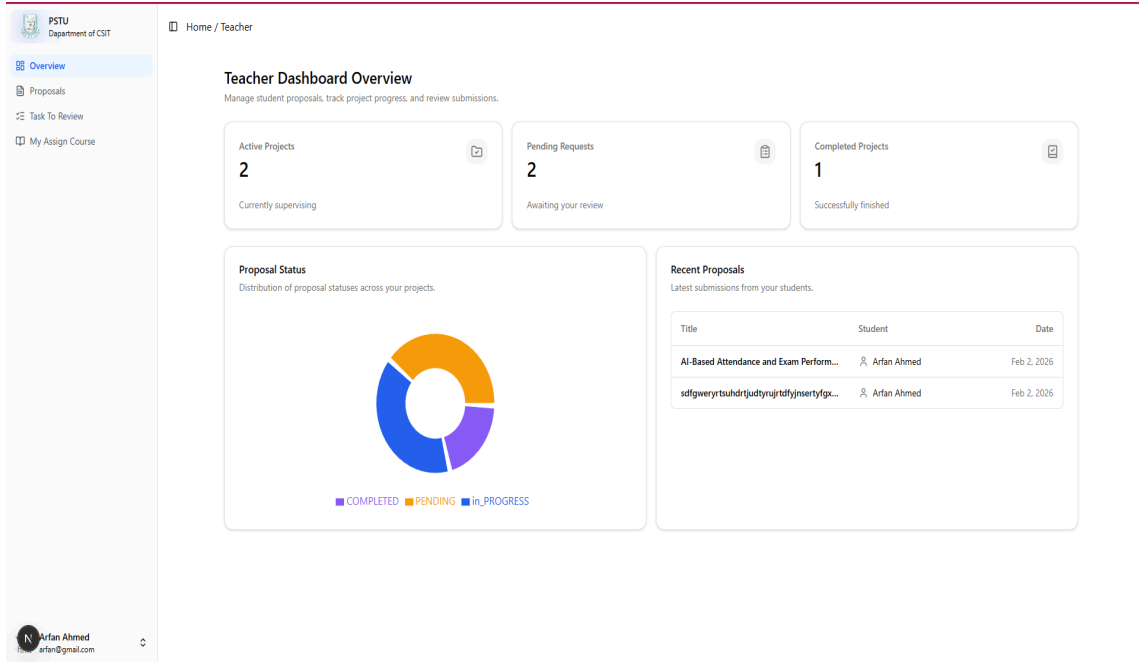


Figure 3: Teacher Admin Panel

5.2.3 Student Dashboard

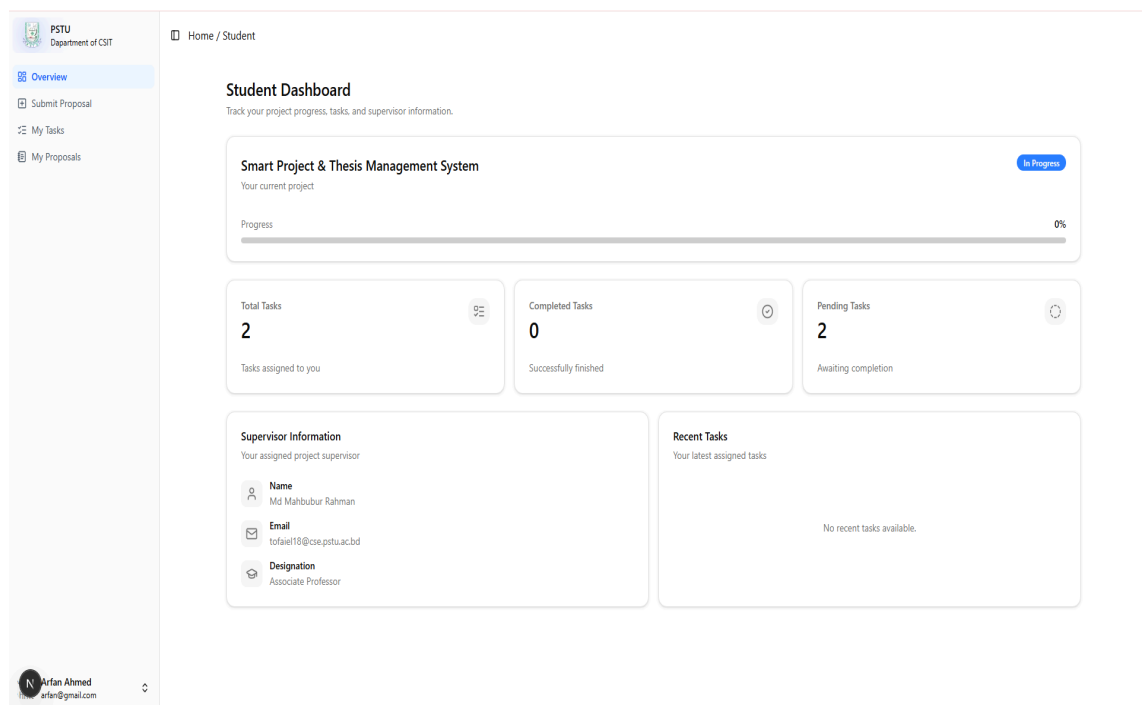


Figure 4: Student Dashboard

5.2.4 Proposal Submission

Submit Proposal

Fill in the details below to submit your project proposal for review

Project Title *

Enter your project title

Abstract *

Provide a brief summary of your project (200-300 words)

Project Objectives *

List the main objectives of your project (one per line)

Methodology *

Describe your approach and methods for completing this project

Expected Outcomes *

What results or deliverables do you expect from this project?

Technologies & Tools *

e.g., React, Node.js, MongoDB, TensorFlow, etc.

Estimated Timeline *

e.g., 6 months, 1 semester, etc.

Type *

Type

Course *

Course

Supervisor *

Select Supervisor

Attachments

Upload supporting documents

Supported formats: PDF, DOC, DOCX, JPG, PNG (Max 3MB each)

Choose Files

Submit Proposal

Need help? Contact your [supervisor](#)

Figure 5: Proposal Submission

5.2.5 My Submissions

PSTU
Department of CSIT

Overview

Submit Proposal

My Tasks

My Proposals

Arfan Ahmed
arfan@gmail.com

Home / Student / My Proposals

My Proposals

Download Report

Search proposals...

Filter by type

Filter by course

Filter by status

Name (A → Z)

AI-Based Attendance and Exam Performance Tracker

THESIS

Course Code: CIT-111
Course Title: Programming Language
Supervisor: Md Mahbubur Rahman
Submitted: 2/2/2026
Updated: 2/2/2026

View Full Details Download Proposal

IoT-Based Smart Energy Monitoring System

THESIS

Course Code: CCE-224
Course Title: Database System Sessional
Supervisor: Dr. Md. Samsuzzaman
Submitted: 2/2/2026
Updated: 2/2/2026

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sdfgwerjtsuhdrjudyrujtdfjnsertyfgdth

THESIS

Course Code: CIT-111
Course Title: Programming Language
Supervisor: Md Mahbubur Rahman
Submitted: 2/2/2026
Updated: 2/2/2026

View Full Details Download Proposal

AI-Based Student Performance Prediction System

PROJECT

Course Code: CIT-311
Course Title: Microprocessors And Assembly Language
Supervisor: Md Mahbubur Rahman
Submitted: 1/31/2026
Updated: 2/1/2026

View Full Details Download Proposal

Food Availability and Online Ordering System for Students

PROJECT

Course Code: CCE-221
Course Title: Digital Logic Design
Supervisor: Md Mahbubur Rahman
Submitted: 1/31/2026
Updated: 1/31/2026

View Full Details Download Proposal

Smart Project & Thesis Management System

PROJECT

Course Code: CIT-111
Course Title: Programming Language
Supervisor: Md Mahbubur Rahman
Submitted: 1/31/2026
Updated: 2/2/2026

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1

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Figure 6: My Submissions

5.2.6 My Tasks

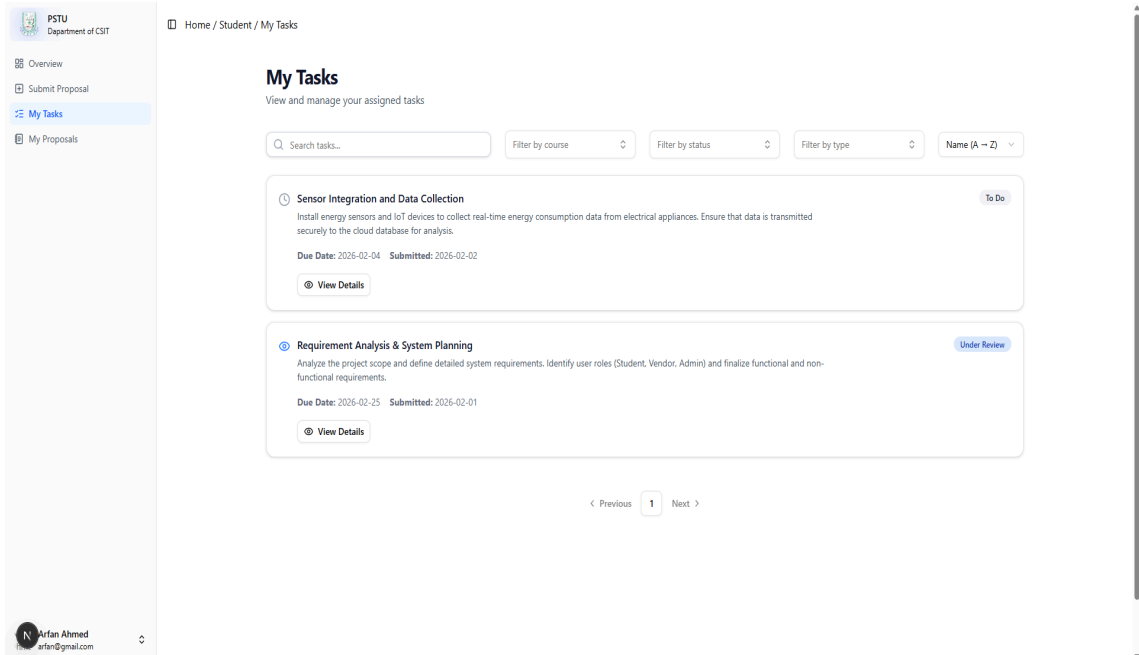


Figure 7: My Tasks

5.2.7 Submitted Proposals

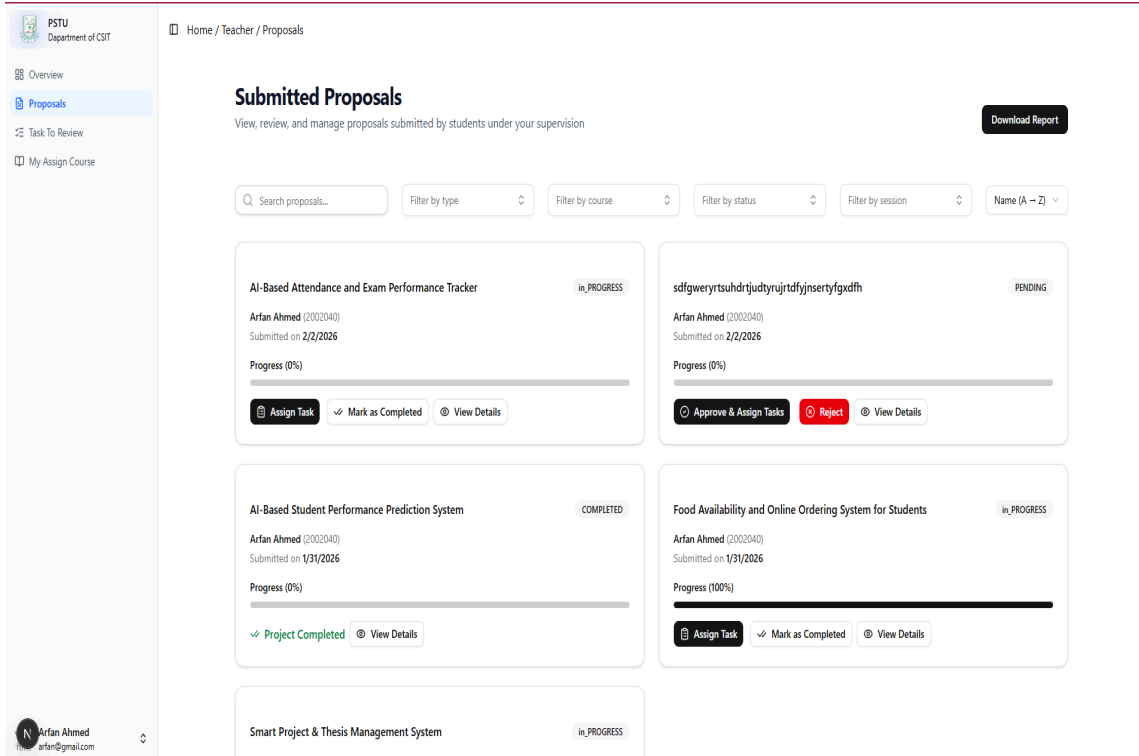


Figure 84: Submitted Proposals.

5.2.8 Task Review

Submitted Proposals
View, review, and manage proposals submitted by students under your supervision

Download Report

Search proposals... Filter by type Filter by course Filter by status Filter by session Name (A → Z)

Proposal Title	Status	Submitted on	Progress	Actions
AI-Based Attendance and Exam Performance Tracker	in_PROGRESS	2/2/2026	0%	Assign Task, Mark as Completed, View Details
sdfgwerjtsuhdrtjudyrtjtdfjynsertyfgxdth	PENDING	2/2/2026	0%	Approve & Assign Tasks, Reject, View Details
AI-Based Student Performance Prediction System	COMPLETED	1/31/2026	0%	Project Completed, View Details
Food Availability and Online Ordering System for Students	in_PROGRESS	1/31/2026	100%	Assign Task, Mark as Completed, View Details
Smart Project & Thesis Management System	in_PROGRESS			

Figure9: Task Review

5.2.9 My Assigned Courses

My Assigned Courses
5 Total Courses

Search assigned courses courseCode or title... Filter by semester Name (A → Z)

Course Title	Semester	Credits	Status
Artificial Intelligence Sessional (CIT-316)	5th	1.5 credits	Active
Microprocessors And Assembly Language (CIT-311)	5th	3 credits	Active
Digital Logic Design (CCE-221)	4th	3 credits	Active
Data Structure And Algorithms (CIT-212)	3rd	1.5 credits	Active
Programming Language (CIT-111)			Active

Figure 10: My Assigned Courses

5.2.10 Project and Thesis Collection

PSTU
Department of CSIT

- Overview
- Project/Thesis Collections**
- Add Course
- Manage Courses
- Add Teacher
- Manage Teachers
- Manage Students
- Manage Admins

Home / Admin / Project Thesis Collections

Submitted Proposals

View, review, and manage proposals submitted by students under your supervision

[Download Report](#)

Filter by type ▾
Filter by course ▾
Filter by status ▾
Filter by session ▾
Name (A – Z) ▾

AI-Based Attendance and Exam Performance Tracker
in_PROGRESS

Arfan Ahmed (2002040)

Submitted on **2/2/2026**

Progress (0%)

[View Details](#)

IoT-Based Smart Energy Monitoring System
PENDING

Arfan Ahmed (2002040)

Submitted on **2/2/2026**

Progress (0%)

[View Details](#)

sdfgwerjtsuhdrtjdtyrujntdyfnsertyfgadfh
PENDING

Arfan Ahmed (2002040)

Submitted on **2/2/2026**

Progress (0%)

[View Details](#)

AI-Based Student Performance Prediction System
COMPLETED

Arfan Ahmed (2002040)

Submitted on **1/31/2026**

Progress (0%)

[View Details](#)

Food Availability and Online Ordering System for Students
in_PROGRESS

Smart Project & Thesis Management System

in_PROGRESS

Figure 11: Project and Thesis Collection

5.2.11 Manage Courses



PSTU

Department of CSIT

Home / Admin / Manage Courses

Overview

Project/Thesis Collections

Add Course

Manage Courses

Add Teacher

Manage Teachers

Manage Students

Manage Admins

Course Management

Manage and organize all courses in your system

Search courses...

Filter by semester

Filter by status

Name (A - Z)

Course Code	Course Title	Credits	Semester	Status	Actions
CSE-420	Project/Thesis	3	8th	ACTIVE	View Edit Delete Share
CCE-411	Algorithm Engineering	3	7th	ACTIVE	View Edit Delete Share
CCE-413	VLSI Design	3	7th	ACTIVE	View Edit Delete Share
CSE-412	Industrial Training	1	7th	ACTIVE	View Edit Delete Share
CSE-410	Project/Thesis	3	7th	ACTIVE	View Edit Delete Share
CIT-324	System and Simulation Sessional	1.5	6th	ACTIVE	View Edit Delete Share
CIT-323	System and Simulation	3	6th	ACTIVE	View Edit Delete Share
CIT-322	Operating System Sessional	1.5	6th	ACTIVE	View Edit Delete Share
CIT-321	Operating System	3	8th	ACTIVE	View Edit Delete Share
CIT-316	Artificial Intelligence Sessional	1.5	5th	ACTIVE	View Edit Delete Share

Arfan Ahmed

arfan@gmail.com

Previous

1

2

3

Next

Figure 12: Manage Courses

5.2.12 Manage Teachers

**PSTU**
Department of CSIT

Overview

Project/Thesis Collections

Add Course

Manage Courses

Add Teacher

Manage Teachers

Manage Students

Manage Admins

Arfan Ahmed

arfan@gmail.com

Home / Admin / Manage Teachers

Manage Teachers

View and manage all registered teachers

Filter by designation

Filter by department

Filter by status

Name (A → Z)

Download Report

Name	Email	Phone	Designation	Department	Status	Actions
Dt. Md. Samsuzzaman	shovon1023456789@gmail.com	01707490998	Professor	Computer Science and Communication Engineering	ACTIVE	  
Professor Chinmay Bepery	abhinash@cse.pstu.ac.bd	01707490998	Professor	Computer Science and Information Technology	ACTIVE	  
Md Mahbubur Rahman	tofaie18@cse.pstu.ac.bd	01707490998	Associate Professor	Computer Science and Information Technology	ACTIVE	  

< Previous 1 Next >

Figure 13: Manage Teachers

5.2.13 Manage Students

**PSTU**
Department of CSIT

Overview

Project/Thesis Collections

Add Course

Manage Courses

Add Teacher

Manage Teachers

Manage Students

Manage Admins

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arfan@gmail.com

Home / Admin / Manage Students

Manage Students

View and manage all registered students in the system

Filter by session

Filter by status

Filter by approval

Name (A → Z)

Download Report

Student	Student ID	Registration No.	Contact	Session	Status	Approval	Account Status	Actions
 Buffy Conner ambiamithila991@gmail.com	2002041	09558	01762915294 Pariatur Noodrum qu	2020-21	ACTIVE		Active	...
 Arfan Ahmed anomious31@gmail.com	2002040	09557	01712345678 Dhannordi, Dhaka	2020-21	ACTIVE		Active	...

< Previous 1 Next >

Figure 14: Mange Students

5.2.14 Manage Admins

The screenshot displays the 'Manage Admins' page. On the left is a sidebar with the following menu items: Overview, Project/Thesis Collections, Add Course, Manage Courses, Add Teacher, Manage Teachers, Manage Students, and Manage Admins (highlighted). The main content area has a breadcrumb 'Home / Admin / Manage Admins' and a title 'Manage Admins' with the subtitle 'View and manage all registered administrators'. A '+ Create Admin' button is in the top right. Below the title is a search bar 'Search admins...' and two filters: 'Filter by status' and 'Name (A - Z)'. A table lists the administrators:

Admin	Email	Phone	Status	Created	Actions
Arfan Ahmed Shohag	arfan18@cse.pstu.ac.bd	+8801700000000	Active	2026-01-24	

At the bottom of the table are pagination controls: '< Previous', '1', and 'Next >'.

Figure 15: Manage Admins

5.3 Summary

This chapter presented the overall software design of the Project and Thesis Automation System through its main interfaces, including the Admin Panel, Teacher Admin Panel, and Student Dashboard. It also covered the core student-facing workflows, such as proposal submission, viewing submitted proposals, and managing assigned tasks, as well as the teacher-facing workflows for reviewing proposals, assigning tasks, and viewing assigned courses. On the administrative side, the chapter showed the interfaces for managing the project and thesis collection, courses, teachers, students, and admin accounts. Together, these designs demonstrate how the system's requirements were translated into a functional and role-based interface for each type of user.

CHAPTER 6

TESTING AND SECURITY

6.1 Introduction

The testing and security are very important and vital for the Project and Thesis Automation System. The testing checks the work and the functionality through the unit and the integration and the system tests so it makes sure the requirements are met and followed. Security testing keeps things safe and guards against people who are not allowed and unauthorized access and also data breaches by using things like the vulnerability assessments and encryption. The plans for the maintenance are very important and crucial for keeping the functionality and the security working.

6.2 Research Objective & Development Approach of Agile Methods

In this chapter we look at the topic of testing and security for our project. The way we did the testing was to give answers to the big questions about how reliable the system is and the data integrity. First what ways of testing are the best and most effective for the academic management systems? Second how do we check and formally verify the requirements we got in the analysis phase? Because we used the Waterfall model we can see that the testing part comes after the implementation part in a sequence. One big part of testing is using the big test cases to make sure all the parts of the software are checked and validated against the first requirements. Also the plan for testing the Project and Thesis Automation System looks at many characteristics:

- Unit Testing this means testing the single parts and the functions of the system to make sure they work right alone and in isolation.
- Integration Testing this means checking that the different parts and modules of the system work together like they should when they are put together and combined.
- System Testing: this means doing tests on the whole and finished system to see if the system follows the rules and specified requirements.
- User Acceptance Testing (UAT): This means the final users, like the faculty and the students, check it to make sure the system does what they need in the real world before it is finished and deployed.

6.2.1 Test Automation

Test automation is the process of making the test cases automatic and running them by using automation tools or software. There are several steps and phases involved in test automation that

help ensure the quality of the software product, and these steps make sure the product is good and high quality. These phases include:

- **Test Planning:** This phase involves making a test plan that shows the test goals and the scope, and the requirements. It also includes finding the types of tests to be done and setting up the test environment and picking the right testing tools for the job.
- **Test Development:** In this phase the test cases are made based on the needs and the specs of the system so the test cases are created using automation tools and are built to be done again and again many times.
- **Test Execution:** Once the test cases are made they are run using the automation tool and the tool acts like a human person who tests and runs the test cases. The results of the test run are then looked at and compared to the results we expected so we can see if the software does what it is supposed to do and meets the requirements.

6.2.2 Regression Tests

Regression testing in the Project and Thesis Automation System checks that the old features still work after the system has been changed and modified. It involves doing the functional and nonfunctional tests again to make sure the system still performs right even though things changed in the system or the environment and it checks that things are still the same and have not broken.

6.3 Summary

In this chapter the focus is on the real results and outcomes of the Project and Thesis Automation System. During the testing we found different bugs and defects which helped with fixing errors and recovery and this made the software work better and run more smoothly and more efficient. The debugging process was done step by step on all the data including the admin and the supervisor and the student modules so that every part was checked.

CHAPTER 7

CONCLUSION

7.1 Introduction

The conclusion section summarizes the findings and outcomes of the study, and provides recommendations for future work. It is an important section as it helps to tie all the pieces of the research together and highlight the key points that have been made throughout the study.

7.2 Project Outcomes

The Project and Thesis Automation System resulted in the successful development of a comprehensive management system using the Next.js framework. It incorporated essential features such as user authentication, project proposal submission, supervisor allocation, thesis progress tracking, and repository management. Various testing techniques were applied to ensure functionality, reliability, and security. The system was designed with a user-centric approach, focusing on usability and accessibility. Overall, the project outcomes reflect the successful application of software engineering principles and methodologies to develop a robust and user-friendly thesis management system.

7.3 Future Works

Some of the future works of the project are the following:

- Integration of plagiarism detection tools to automatically verify the originality of submitted proposals and theses.
- Automated defense scheduling system to allocate presentation slots based on faculty and student availability.
- Implementation of a notification system (SMS/Email) and development of a mobile application for real-time updates.

7.4 Summary

In conclusion, the Project and Thesis Automation System successfully applied software engineering principles and the Waterfall model to enhance administrative efficiency and research workflow. Future work includes integration of plagiarism detection tools and refining user interfaces for an improved experience. Overall, the project demonstrates effective project management and collaborative development efforts.

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